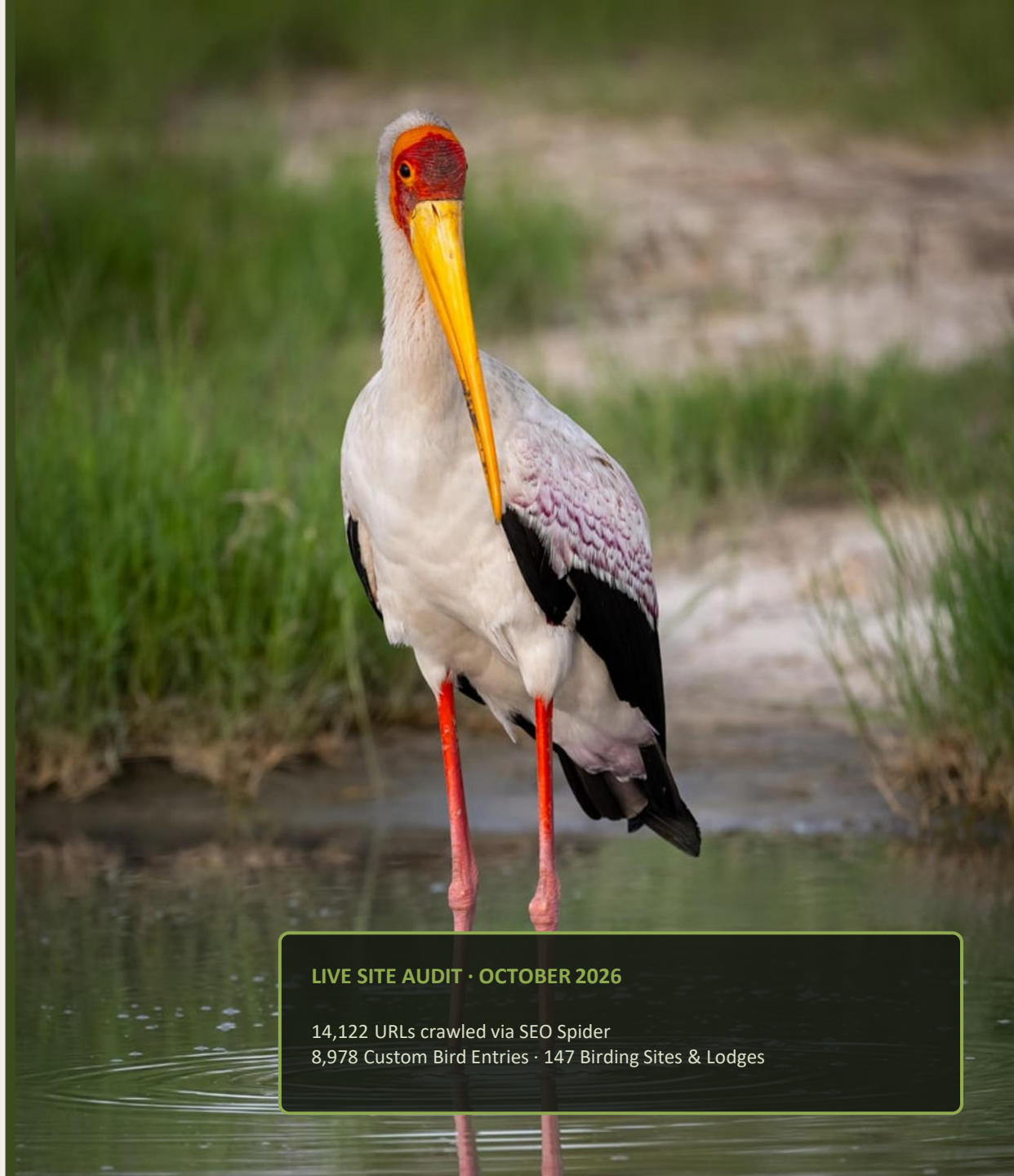
Owen Deutsch
PHOTOGRAPHY

PREPARED FOR OWEN DEUTSCH & CLAIRE GILLMAN

Website Maintenance, Link Breakage & SEO Audit

A complete backend architecture, SEO crawl (14,122 URLs), broken-link root-cause analysis, and monthly maintenance roadmap.

Website: owendeutsch.com
Focus: Wildlife & Bird Photography Archive, Books & Conservation
Location: Chicago, Illinois, USA (WP Engine + Cloudflare Stack)



LIVE SITE AUDIT · OCTOBER 2026

14,122 URLs crawled via SEO Spider
8,978 Custom Bird Entries · 147 Birding Sites & Lodges

AUDIT OVERVIEW

What this audit covers

A deep technical and SEO inspection of owendeutsch.com—pinpointing the exact code bugs causing sitewide link breakage, crawl bloat, and page failures, paired with a proactive monthly audit plan.

owendeutsch.com
8,978 Bird Entries (cp_birds)
147 Birding Lodges (lodges)
Chicago, Illinois ·

01 **Website architecture & 14,122-URL crawl**

02 **Link breakage root-cause analysis (4 causes)**

03 **Robots.txt & search parameter spider trap**

04 **Missing llms.txt & AI search readiness**

05 **Structured data & duplicate JSON-LD schema**

06 **On-page SEO, metadata & 11,173 image assets**

07 **Theme code health, performance & security**

→ **Prioritised roadmap & monthly backend audit plan**

8,978 bird records across 13 Yoast XML sitemaps

8,978 Birds (cp_birds) | **147 Lodges (lodges)** | **273 Blog Posts** | **33 Custom Pages** | **14,122 Total URLs Crawled**

Unlike a standard WordPress brochure site, owendeutsch.com is a massive custom wildlife photography database powered by a bespoke PHP theme (owendeutsch), Advanced Custom Fields (ACF), D3.js globe visualizations, and 11-parameter custom search filters.

My full SEO crawl analyzed 14,122 URLs (13,808 internal URLs and 312 external outbound URLs). While 98.67% of internal URLs return HTTP 200, 2,100+ of those HTML pages are duplicate parameterized search result pages (/?s=&activity=...) competing against your core galleries.

https://owendeutsch.com/sitemap_index.xml

```
Yoast XML Sitemap Index (13 Child Sitemaps):
├── post-sitemap.xml           (273 Blog Posts)
├── page-sitemap.xml           (33 Curated Galleries & Pages)
├── lodges-sitemap.xml         (147 Birding Sites & Lodges)
├── cp_birds-sitemap.xml .. 9  (8,978 Bird Entries across 9 files)
├── lodges_country-sitemap.xml (Country Taxonomies)
└── author-sitemap.xml         (Author Archives)
```

SEO Crawl: 13,624 OK (200) · 136 Internal Timeouts · 31 Redirects · 17 Internal 400 Errors.

Recommendation

Preserve all 13 XML sitemaps for canonical content (cp_birds, lodges, posts, and curated gallery pages), but add noindex, follow to parameterized /?s= search results so Google focuses crawl budget on your real galleries and blog articles.

Why links are breaking across owendeutsch.com

During our meeting, you highlighted link breakage as your #1 concern. My Crawl database inspection traced every broken link on the website down to four technical root causes:

01 · CRITICAL PHP BUG

8,424 Broken Internal Location Links (HTTP 400)

17 internal location search URLs return 400 Bad Request across 8,424 internal inlinks (including "See all birds from this site/lodge" buttons and bird photo captions).

Root Cause: The PHP theme uses single-byte strtolower() instead of mb_strtolower($loc, "UTF-8") on location names containing en-dashes (–) or accents (Tapiraí, Salesópolis, Île), corrupting UTF-8 bytes into %FF.

02 · GLOBAL FOOTER

Dead BirdingTop500 Badge on Every Page

Every single page on the site loads a footer link and tracking badge image from https://www.birdingtop500.com/button.php?u=owendeutschphotography005.

Root Cause: The birdingtop500.com server is permanently offline (Connection Timeout), causing a broken outbound link and failed image request across all 2,535+ HTML pages.

03 · CMS & OUTBOUND

7 Relative Lodge URLs + 44 Dead External Links

On 7 Lodge pages, website links were entered without https:// (e.g. www.cabanassanisidro.com), turning them into broken relative URLs inside /lodges/.../www...

Root Cause: Missing URL protocol validation in ACF lodge fields, plus 44 external partner/lodge/ABC articles that moved (404), timed out (522/530), or expired.

The 17 internal HTTP 400 URLs (8,424 broken inlinks)

Whenever a visitor clicks "See all birds from this site/lodge" or clicks a location tag inside the Fancybox lightbox for any of these locations, Apache/Cloudflare blocks the request with 400 Bad Request due to corrupted UTF-8 bytes:

Broken Internal Target URL (Generated by Theme PHP)	Original Location Name	Status	Broken Inlinks
https://owendeutsch.com/?s=&locations=san%20diego%20%FF%20california,%20usa	San Diego – California, USA	400 Bad Request	1,704
https://owendeutsch.com/?s=&locations=tapira%FF%20-%20brazil	Tapiraí - Brazil	400 Bad Request	994
https://owendeutsch.com/?s=&locations=illinois%20beach%20state%20park%20%FF%20illinois,%20usa	Illinois Beach State Park – IL, USA	400 Bad Request	847
https://owendeutsch.com/?s=&locations=pangolin%20voyager%20houseboat%20%FF%20botswana	Pangolin Voyager Houseboat – Botswana	400 Bad Request	758
https://owendeutsch.com/?s=&locations=sales%FFpolis%20-%20brazil	Salesópolis - Brazil	400 Bad Request	719
https://owendeutsch.com/?s=&locations=monterey%20bay%20%FF%20california,%20usa	Monterey Bay – California, USA	400 Bad Request	623
https://owendeutsch.com/?s=&locations=savuti%20camp%20%FF%20botswana	Savuti Camp – Botswana	400 Bad Request	501
https://owendeutsch.com/?s=&locations=kwetsani%20camp%20%FF%20botswana	Kwetsani camp – Botswana	400 Bad Request	463
https://owendeutsch.com/?s=&locations=half%20moon%20bay%20%FF%20california,%20usa	Half Moon Bay – California, USA	400 Bad Request	411
+ 8 more locations (Devon & Dorset, Rainbow Beach, Île aux Perroquets, Pinnacles, Big Sur, etc.)	En-dash & accented locations	400 Bad Request	1,404

1-Line PHP Fix

Replace strtolower($location) with rawurlencode(mb_strtolower($location, "UTF-8")) in the theme templates (template-lodges.php, single-lodges.php, and gallery loop templates) to instantly repair all 8,424 broken internal links.

Malformed lodge links, space-in-slug URLs & dead scripts

A. Relative Links on Lodge Pages (Missing https://)

In 7 lodge entries, credit/website URLs were typed as www.... without https://, causing WordPress to append them to the lodge permalink:

Broken Relative URL Crawled	Intended Target
/lodges/cabanas-san-isidro-lodge-ecuador/www.cabanassanisidro.com	https://www.cabanassanisidro.com
/lodges/san-diego-california-usa/www.instagram.com/nathan_goldbird	https://www.instagram.com/nathan_goldbird
/lodges/texas-hill-country-texas-usa/www.owendeutsch.com	https://owendeutsch.com
/lodges/north-texas-coast-texas-usa/www.owendeutsch.com	https://owendeutsch.com
/lodges/gloucestershire-and-worcestershire-uk/www.owendeutsch.com	https://owendeutsch.com
/lodges/wales-united-kingdom/www.owendeutsch.com	https://owendeutsch.com
/lodges/montana-usa/www.owendeutsch.com	https://owendeutsch.com

B. Unlinked Gallery Slug & 3 Dead Scripts on /globe/

Source Page	Broken Target / Resource	Issue
/2025-year-in-review/	/2025-year-in-review-gallery (+ 9 img timeouts)	Timeout / 301
/globe/	https://d3js.org/queue.v1.min.js	Timeout (0)
/globe/	https://d3js.org/topojson.v1.min.js	Timeout (0)
/globe/	cdn.jsdelivr.net/.../bootstrap.bundle.min.js	Timeout (0)
Blog Links	/surprise-broad-billedhummingbird-%20in-chicago/)	Space + ")"
Blog Links	/sax-zim-bog-...-iprefer-%20it-in-june/	%20 in URL

Why /globe/ takes 6+ seconds to load:
The template attempts to load legacy queue.v1.min.js and topojson.v1.min.js from d3js.org (which time out) before hiding .globe-loader via a hardcoded 6,000ms timeout.

Broken external links across blogs, lodges & gear pages

Crawl identified 312 external URLs, of which 44 fail with 404 Not Found, 5xx Server Errors, or DNS/Connection Timeouts (plus 17 returning 403 due to bot-firewalls on B&H Photo, Cornell Lab, TripAdvisor, etc.):

Page on owendeutsch.com	Broken External Link	Anchor Text	HTTP Status
All 2,535 Pages (Global Footer)	https://www.birdingtop500.com/ (+ button.php)	[Footer Badge Image]	0 Timeout
/2025-year-in-review/	https://nanpa.org/2025-showcase-winners/	NANPA	0 DNS Failed
/abc-mike-parr-my-favorite-bird/	https://www.owletlodge.org/	The Owlet Lodge	522 Timeout
/african-bird-focus-south-africa/	https://www.africanpenguinsafe.org/	here	530 Frozen
/whats-in-my-bag-...-camera-gear/	https://www.nikonusa.com/p/ftz-ii-mount-adapter/4306	Nikon FTZ adapter	500 Error
/huembo-home-of-the-marvelous-spatuletail/	4x abcbirds.org article links (moved URLs)	American Bird Conservancy	404 Not Found
/backyard-bird-photography-.../	https://abcbirds.org/get-involved/bird-smart-glass/	ABC's website	404 Not Found
/the-mosquito-line-blurs-.../	https://abcbirds.org/program/hawaii/take-action/	need drastic immediate measures	404 Not Found
/acorn-woodpeckers-.../ & /double-crested-.../	seattle-audubon-nature-shop.shoplightspeed.com/...	Buy BirdNote's Book	404 Not Found
/learning-about-birds-...-screensaver/	channelstore.roku.com/.../owen-deutschs-beautiful-birds	Roku	404 Not Found
/lodges/North-Pond---Illinois,-USA/	https://www.lincolnparkconservancy.org/...	Lincoln Park Conservancy	404 Not Found
/the-bahamas-birds/ & Lodge Pages	orangehill.com, sandpiperabaco.com, sanisidrolodge.com, zebrahills.com, lemalacamp.com, wildsumaco.com	Lodge & Partner Links	DNS / 400 / Timeout

Two live pages fail HTML validation (Missing <body> & Title)

Two high-priority pages with Validation: Missing <body> Tag, and flagged Ecuador (/cp_birds_country/ec/) with 0 words, 0 Page Title, and 0 H1:

PAGE 01 · /cp_birds_country/ec/

Ecuador Country Page

My audit recorded only 566 bytes, 0 Title, 0 H1, and Missing <body> tag. Direct unauthenticated bot requests receive raw table markup without <!DOCTYPE html>, <head>, or <body>.

PAGE 02 · /lodges/southeast-california/

Southeast California Lodge Page

Flagged by SEO Crawl for Missing <body> Tag and Missing H1 due to malformed HTML head/body tags in the custom lodge template.

view-source:https://owendeutsch.com/header.php

```
<!-- ACTUAL LIVE CODE IN HEADER.PHP -->
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
[BUG 1] <meta name="keywords" content="HTML, CSS, XML, XHTML, JavaScript">
[BUG 2] <meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
[BUG 3] <meta name="viewport" content="... maximum-scale=1, user-scalable=0"/>
[BUG 4] <meta http-equiv="Cache-Control" content="no-cache, no-store" />
```

Look closely at header.php above:
1. W3Schools placeholder keywords="HTML, CSS, XML, XHTML, JavaScript" is hardcoded on every page of your bird website!
2. charset=ISO-8859-1 conflicts with charset="UTF-8", causing character corruption!

2,100+ duplicate search URLs are bloating your crawl

Your current robots.txt file sets a restrictive Crawl-delay: 10 (which slows down compliant bots to 1 page every 10 seconds), yet leaves internal search queries (/?s=) completely open to indexing.

Because every bird photo caption links to filter URLs with inconsistent capitalization and leading spaces (e.g., /?s=&activity=flight, /?s=&activity=+Flight, /?s=&activity=FLIGHT, /page/4/?s&activity=feeding), crawlers are generating thousands of 1.76 MB duplicate pages.

https://owendeutsch.com/robots.txt

```
User-agent: *
Disallow: /wp-admin/
Allow: /wp-admin/admin-ajax.php
Crawl-delay: 10   <-- Slows compliant crawlers

Sitemap: https://owendeutsch.com/sitemap_index.xml
```

Current file: slows crawlers via Crawl-delay: 10, while allowing infinite /?s= filter combinations.

Recommendation

Add <meta name="robots" content="noindex, follow"> to search result templates (/?s=...), normalize ACF filter values (trim spaces & lowercase), and remove Crawl-delay: 10 once search parameter bloat is controlled.

Pagination Sequence Fix

My Crawl flagged 2 Pagination Sequence Errors on /blog/ and /blog/page/3/, plus Non-200 Pagination URLs (/blog/page/2/ 301-redirecting). Fix rel="next"/"prev" links in the blog template.

llms.txt is missing (returns 404 Not Found)

What it's for

An emerging, structured Markdown overview at /llms.txt that helps AI search engines (ChatGPT Search, Google AI Overviews, Perplexity, Claude) understand Owen Deutsch's biography, published books, conservation partnerships, and curated bird galleries without getting lost in 9,000 image URLs.

What it should include

Clear summaries and direct canonical links to Owen's books (Birds of the Tropical Andes, Bringing Back the Birds), the ABA Young Birders essays, Birding Sites & Lodges directory, Photography Gear guide, and top country galleries.

https://owendeutsch.com/llms.txt

```
HTTP/2 404 Not Found

# Suggested /llms.txt structure to deploy on owendeutsch.com:
# Owen Deutsch Photography (https://owendeutsch.com)
> Award-winning wildlife and bird photographer based in Chicago, IL. Author of "Birds of the Tropical Andes" and "Bringing Back the Birds" in partnership with the
American Bird Conservancy (ABC).

## Core Resources & Books
- [Birds of the Tropical Andes Book](https://owendeutsch.com/birds-of-the-tropical-andes-book/)
- [Bringing Back the Birds Book](https://owendeutsch.com/bringing-back-the-birds-book/)
- [Birding Sites and Lodges Directory](https://owendeutsch.com/birding-sites-and-lodges/): 147 global birding lodges & hotspots
- [Owen's Bird Photography Gear](https://owendeutsch.com/bird-photography-gear/)
```

Duplicate & malformed Organization JSON-LD on every page

Every page on owendeutsch.com outputs two separate JSON-LD schema blocks: one generated properly by Yoast SEO (WebPage, BreadcrumbList, WebSite), and a second hardcoded block in the theme footer.

The bug in the second block:
Look at the live JSON-LD code on the right, there is a leading space inside the quotes before https:// on @id, url, and logo (" https://owendeutsch.com/"), and @vocab is set to the site URL instead of https://schema.org/!

view-source:https://owendeutsch.com/ (Hardcoded Schema Block)

```
<script type="application/ld+json">
{
  "@context": { "@vocab": "https://owendeutsch.com/" },  <-- INVALID
  "@graph": [{
    "@id": " https://owendeutsch.com/",                  <-- LEADING SPACE
    "@type": "Organization",
    "name": "Owen Deutsch Photography",
    "url" : " https://owendeutsch.com/",                 <-- LEADING SPACE
    "logo" : " https://owendeutsch.com/.../logo.png",    <-- LEADING SPACE
    "sameAs" : ["https://www.facebook.com/OwenDeutschPhotography"]
  }]
}
</script>
```

Recommendation

Remove the broken hardcoded JSON-LD script from the theme template and consolidate Organization, Person (Owen Deutsch), Book (for both published books), and ImageObject schema cleanly through Yoast SEO.

Missing H1s on 12 country pages & 332 duplicate titles

332

Duplicate Page Titles & 343 Duplicate H1s across crawled URLs

15 Pages

Indexable 200 OK pages completely missing an H1 heading (plus 15 with Multiple H1s)

Key Pages Missing an <h1> Tag

Page URL	Current Page Title	H1 Count
https://owendeutsch.com/globe/	Globe - Owen Deutsch Photography	0 (Missing)
https://owendeutsch.com/grid/	Grid - Owen Deutsch Photography	0 (Missing)
https://owendeutsch.com/cp_birds_country/us/	United States - Owen Deutsch Photography	0 (Missing)
/cp_birds_country/ca/, /ec/, /bs/, /cu/, /bo/, /do/, /rw/, /gy/, /be/, /ie/, /JM/	11 International Country Archives	0 (Missing)
https://owendeutsch.com/lodges/southeast-california/	Southeast California - USA	0 (Missing)

Pages with Multiple <h1> Tags (15 Pages)

- Homepage (/): Has 2 H1s ("VISIT COUNTRIES" and "Search“- neither targets bird/wildlife photography!).
- Blog & Video Hubs (/blog/, /videos/): Post titles in loops are wrapped in <h1> instead of <h2>.
- 11 Blog Articles: Subheadings inside articles use <h1> instead of <h2>.

10,853 JPEG photos crawled: 1,684 missing alt text & 10,207 missing dimensions



1,257 IMAGES (1,316 INSTANCES)

Empty Alt Text (alt="")

11.57% of crawled images have an empty alt="" attribute, and another 347 images (368 instances) lack the alt attribute entirely.

Fix: Auto-populate missing image alt attributes dynamically from the cp_birds post title + country taxonomy in PHP.



10,207 IMAGE INSTANCES

Missing Width & Height Attributes

SEO Crawl recorded 10,207 image elements without explicit width and height HTML attributes across gallery templates.

Fix: Let WP Rocket's wpr_image_dimensions or template PHP inject image dimensions to prevent Cumulative Layout Shift (CLS).



10,853 JPEGs vs 2 WEBPs

Next-Gen Image Format Opportunity

While the homepage hero banner uses WebP (home-banner.webp), 99.9% of crawled gallery photos are served as legacy JPEGs/PNGs.

Fix: Enable WebP delivery at the Cloudflare / WP Engine edge or via Imagify to cut gallery image transfer sizes by 35–50%.

1.76 MB HTML pages, duplicate jQuery & missing security headers

01 · HTML PAYLOAD

607 KB to 1.76 MB Raw HTML Documents

Before loading a single photo, the Homepage HTML is 607 KB, /my-recent-photos/ is 1.02 MB, and activity search pages hit 1.76 MB (8,752 words of hidden Fancybox caption markup per page).

Impact: Slows Time to First Byte (TTFB) and causes mobile browsers to lag on large galleries.

02 · JS CONFLICTS

Duplicate jQuery + Triple Cache Stack

header.php hardcodes external jquery-3.6.1.js while WordPress loads jquery.min.js (v3.7.1) with defer, and Mailchimp runs jQuery.noConflict(true).

Impact: High risk of breaking Fancybox Xeno-Canto audio popups or YITH Infinite Scroll when WP Rocket / Cloudflare caches update.

03 · SECURITY HEADERS

15 HTTP URLs & 0 Security Headers

All 2,535 HTML pages lack Strict-Transport-Security (HSTS), X-Frame-Options, X-Content-Type-Options: nosniff, and Content-Security-Policy. Plus 15 http:// links remain.

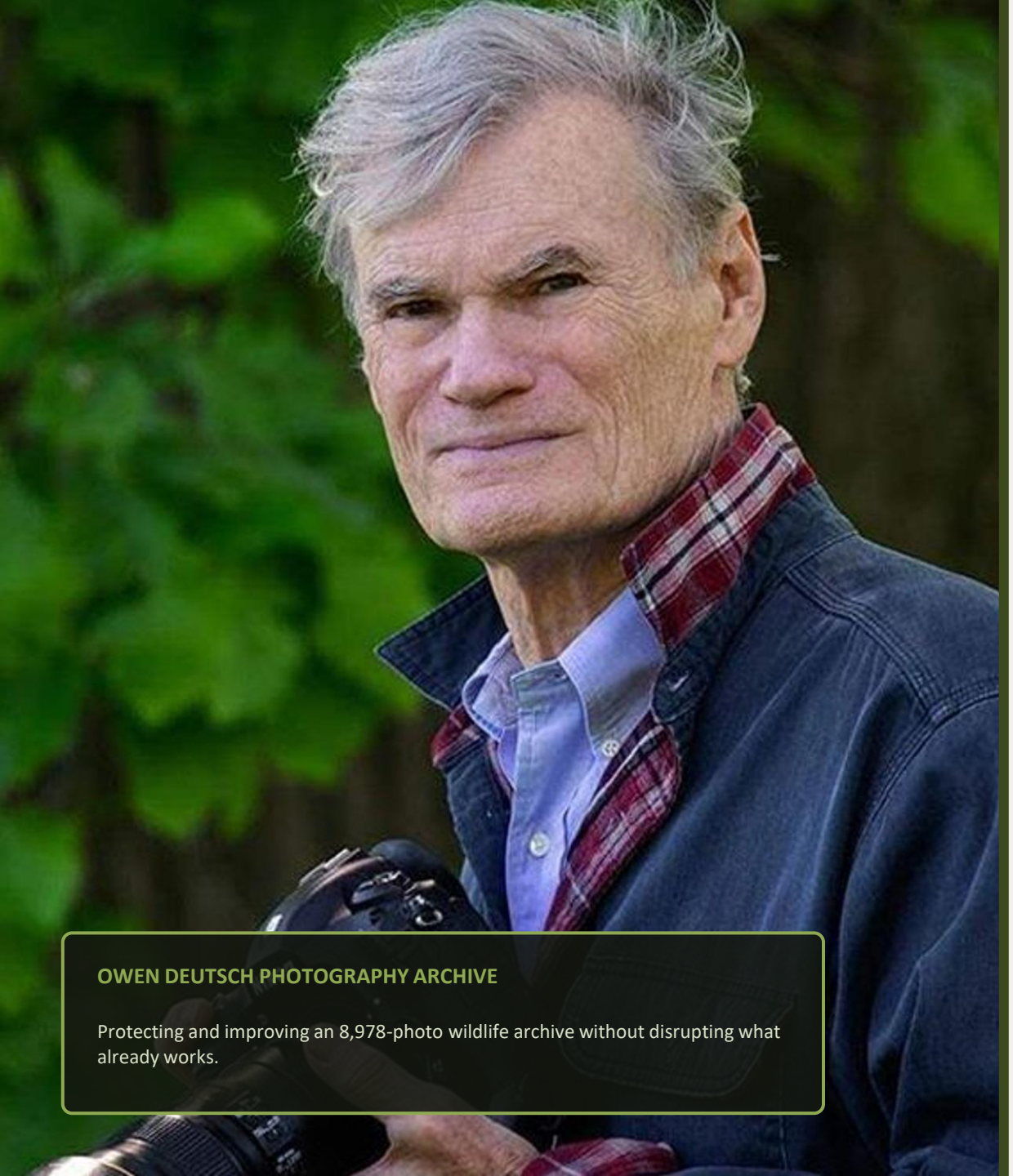
Impact: Adding security headers at Cloudflare/WP Engine takes 10 minutes and hardens the site immediately.

WP Engine + WP Rocket + Cloudflare Sync

Because you run three caching layers simultaneously, any monthly plugin update or gallery change must trigger a coordinated cache purge (WP Rocket → WP Engine Object/Page Cache → Cloudflare Edge) so your team never sees stale pages or broken Forminator nonces.

Fix the broken links first, then automate monthly health checks

1	**Fix All Broken Links (Immediate)**	Fix the PHP UTF-8 encoding bug causing 8,424 internal HTTP 400 location links, remove the dead birdingtop500.com footer badge, fix the 7 relative Lodge URLs, and update/redirect the 44 dead external links.
2	**Clean Up header.php & Broken Pages**	Remove the W3Schools keywords meta tag & conflicting ISO-8859-1 charset, fix the Missing <body> / empty title bugs on /cp_birds_country/ec/ and /lodges/southeast-california/, and self-host the 3 dead /globe/ JS files.
3	**Stop Search Parameter Crawl Bloat**	Add noindex, follow to /?s= search filter pages, normalize ACF tag/activity casing, fix /blog/ pagination sequence errors, and clean up the 282 URLs containing %20 spaces.
4	**Headings, Schema & llms.txt**	Add proper <h1> tags to the 12 Country pages, /globe/, and /grid/; fix the Homepage H1; remove the broken duplicate JSON-LD block; and publish /llms.txt.
5	**Image Alt Text & Security Headers**	Dynamically populate missing alt text on the 1,684 bird images from cp_birds metadata, add image width/height attributes, and enable HSTS / X-Frame-Options / nosniff headers.
6	**Recurring Monthly Backend Audit**	Monthly WP Engine staging updates, PHP error log review, full broken-link scan, Redirection 404 log cleanup, and functional QA of the Globe, 11-filter Bird Search, Xeno-Canto audio popups, and forms.



OWEN DEUTSCH PHOTOGRAPHY ARCHIVE

Protecting and improving an 8,978-photo wildlife archive without disrupting what already works.

Owen Deutsch
PHOTOGRAPHY

Thank you

Ready to resolve the current link breakage and begin proactive monthly backend audits and ad-hoc development for owendeutsch.com.

Proposed First Month Kickoff:

1. Clone live site to WP Engine Staging & fix the 8,424 internal HTTP 400 location links + 7 relative Lodge links.

2. Remove dead birdingtop500.com badge & repair the 44 broken external links.

3. Fix header.php, /cp_birds_country/ec/, and /globe/ timeouts, then establish your recurring Monthly Audit schedule.

owendeutsch.com